Wei Xiong

Member of Technical Staff, OpenAI | RL / Frontier Team
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Industry Experience

OpenAI Jan 2026 - present
Member of Technical Staff, RL / Frontier Team
- Develop reinforcement learning algorithms to advance the capabilities of frontier AI models.

Meta FAIR May 2025 - Aug 2025
Research Scientist Intern, Alignment Team
Hosts: Sainbayar Sukhbaatar and Jason Weston
- Developed a stepwise generative judge trained with RL, combining self-segmentation of reasoning trajectories with a process-reward training recipe.

Google DeepMind May 2024 - Apr 2025
Student Researcher, Gemini Post-Training Team
Hosts: Tianqi Liu and Bilal Piot
- Designed multi-turn DPO for tool-using agents that reason over model-generated tokens and external feedback.
- Investigated connections between process rewards and Q-learning, and scaled training for generative process reward models.

Microsoft Research Asia Jan 2021 - Jun 2021
Research Intern
- Developed RL-based bandwidth estimation methods for real-time communications in Microsoft Teams; studied distributional reinforcement learning.

Education

University of Illinois Urbana-Champaign 2026
Ph.D. in Computer Science
Advisors: Tong Zhang and Nan Jiang

The Hong Kong University of Science and Technology 2023
M.Phil. in Mathematics
Advisor: Tong Zhang

University of Science and Technology of China 2021
B.Sc. in Mathematics & Electronic Engineering

Honors & Fellowships

Google PhD Fellowship 2025

Best Paper Award, NAACL Demo Track 2024

Hong Kong PhD Fellowship Scheme 2021 - 2023

Best Teaching Assistant Award, HKUST 2022, 2023

Outstanding Graduate of Anhui Province and USTC 2021

Guo Moruo Scholarship, Finalist 2020

Selected Research & Open Source

My current work at OpenAI focuses on reinforcement learning algorithms to advance the capabilities of frontier AI models.

During my Ph.D. at UIUC, I studied reinforcement learning and its applications to LLM post-training, spanning theoretical foundations, algorithm design, and practical training methods. I developed:

1. RAFT / 0-1 reward policy gradient.

Rejection-sampling fine-tuning for more stable and efficient RL learning. Our work was among the first to show that such simple policy-gradient methods can outperform PPO on RLHF/RLVR tasks.

Papers: RAFT (TMLR 2023) | A Minimalist Approach to LLM Reasoning (2025).

2. Online DPO and a theoretical framework for KL-regularized contextual bandits.

Papers: Iterative Preference Learning (ICML 2024) | Logarithmic Regret for Online KL-Regularized RL (ICML 2025).

3. Adaptive sampling for Monte Carlo estimation in RAFT and policy gradients:

adaptive sampling budget allocation for variance reduction and learning-signal recovery in policy gradients.

Papers: Gradient Variance Minimization (NeurIPS 2025) | Reinforce-Ada (2025).

Prior to my work on LLMs, I studied the theory of online decision making:

- Learnability and regret analysis in multi-agent Markov games [i] [ii] [iii].
- A unified viewpoint of learnability and generalization in online decision making [iv] [v].
- Improved algorithm design and analysis in decentralized convex optimization [vi].

Open source

I led and contributed to RLHFlow, the open-source framework accompanying our work on online DPO (**2,000+ GitHub stars, 500 academic citations, 1M Hugging Face downloads**), and trained many widely used reward models.

Papers: RLHF Workflow (TMLR 2024) | Multi-Objective Reward Modeling (EMNLP 2024).

Selected Publications

* Equal contribution. Author order follows the cited papers.

Reinforce-Ada: An Adaptive Sampling Framework under Non-linear RL Objectives
Wei Xiong, Chenlu Ye, Baohao Liao, Hanze Dong, Xinxing Xu, Christof Monz, Jiang Bian, Nan Jiang, Tong Zhang
arXiv, 2025 | Paper

Optimizing Chain-of-Thought Reasoners via Gradient Variance Minimization in Rejection Sampling and RL
Jiarui Yao, Yifan Hao, Hanning Zhang, Hanze Dong, **Wei Xiong**, Nan Jiang, Tong Zhang
NeurIPS 2025 | Paper

A Minimalist Approach to LLM Reasoning: from Rejection Sampling to Reinforce
Wei Xiong, Jiarui Yao, Yuhui Xu, Bo Pang, Lei Wang, Doyen Sahoo, Junnan Li, Nan Jiang, Tong Zhang, Caiming Xiong, Hanze Dong
arXiv, 2025 | Paper

Logarithmic Regret for Online KL-Regularized Reinforcement Learning
Heyang Zhao, Chenlu Ye, **Wei Xiong**, Quanquan Gu, Tong Zhang
ICML 2025 | Paper

Self-rewarding Correction for Mathematical Reasoning
Wei Xiong, Hanning Zhang, Chenlu Ye, Lichang Chen, Nan Jiang, Tong Zhang
arXiv, 2025 | Paper

Building Math Agents with Multi-Turn Iterative Preference Learning
Wei Xiong, Chengshuai Shi, Jiaming Shen, Aviv Rosenberg, Zhen Qin, Daniele Calandriello, Misha Khalman, Rishabh Joshi, Bilal Piot, Mohammad Saleh, Chi Jin, Tong Zhang, Tianqi Liu
ICLR 2025 | Paper

Selected Publications (continued)

RLHF Workflow: From Reward Modeling to Online RLHF
Hanze Dong*, **Wei Xiong***, Bo Pang*, Haoxiang Wang*, Han Zhao, Yingbo Zhou, Nan Jiang, Doyen Sahoo, Caiming Xiong, Tong Zhang
Transactions on Machine Learning Research, 2024 | Paper

Interpretable Preferences via Multi-Objective Reward Modeling and Mixture-of-Experts
Haoxiang Wang*, **Wei Xiong***, Tengyang Xie, Han Zhao, Tong Zhang
EMNLP 2024 | Paper

Iterative Preference Learning from Human Feedback: Bridging Theory and Practice for RLHF under KL-Constraint
Wei Xiong*, Hanze Dong*, Chenlu Ye*, Ziqi Wang, Han Zhong, Heng Ji, Nan Jiang, Tong Zhang
ICML 2024 | Paper

Mitigating the Alignment Tax of RLHF
Yong Lin*, Hangyu Lin*, **Wei Xiong***, Shizhe Diao*, Jianmeng Liu, Jipeng Zhang, Rui Pan, Haoxiang Wang, Wenbin Hu, Hanning Zhang, Hanze Dong, Renjie Pi, Han Zhao, Nan Jiang, Yuan Yao, Heng Ji, Tong Zhang
EMNLP 2024 | Paper

A Theoretical Analysis of Nash Learning from Human Feedback under General KL-Regularized Preference
Chenlu Ye, **Wei Xiong**, Yuheng Zhang, Nan Jiang, Tong Zhang
NeurIPS 2024 | Paper

RAFT: Reward rAnked FineTuning for Generative Foundation Model Alignment
Hanze Dong*, **Wei Xiong***, Deepanshu Goyal, Yihan Zhang, Winnie Chow, Rui Pan, Shizhe Diao, Jipeng Zhang, Kashun Shum, Tong Zhang
Transactions on Machine Learning Research, 2023 | Paper

One Objective to Rule Them All: A Maximization Objective Fusing Estimation and Planning for Exploration
Zhihan Liu*, Miao Lu*, **Wei Xiong***, Han Zhong, Hao Hu, Shenao Zhang, Sirui Zheng, Zhuoran Yang, Zhaoran Wang
NeurIPS 2023 | Paper

GEC: A Unified Framework for Interactive Decision Making in MDP, POMDP, and Beyond
Han Zhong*, **Wei Xiong***, Sirui Zheng, Liwei Wang, Zhaoran Wang, Zhuoran Yang, Tong Zhang
Mathematics of Operations Research | Paper